introducing the following topological description function (TDF):

$$\begin{cases} \phi^i(\mathbf{x}) < 0 & \text{if } \mathbf{x} \in \Omega^i \\ \phi^i(\mathbf{x}) = 0 & \text{if } \mathbf{x} \in \partial\Omega^i \\ \phi^i(\mathbf{x}) > 0 & \text{if } \mathbf{x} \in \Omega \setminus \Omega^i \end{cases} \tag{8}$$

For the layered structure, ϕ^i is the TDF of the i-th interface and Ω^i is the region occupied from the curve C^i and C^0. The mathematical expression of TDF for a layered structure can be defined as:

$$\phi^i(\mathbf{x}) = r(t) - r^i(t) \tag{9}$$

where $r^i(t)$ is the distance between the curve C^i and C^0 at any arbitrary point on the domain. To this end, the m-th layer of the structure ($m > 1$) can be defined by $\phi^{m-1}(\mathbf{x}) > 0$ and $\phi^m(\mathbf{x}) \le 0$, except for the first layer $\phi^1(\mathbf{x}) < 0$ and last layer $\phi^n(\mathbf{x}) > 0$. To this end, A bilayer structure can be expressed by the proposed method as shown in Fig. 2. Note that we assume that each layer is homogeneous and did not consider spatial variations within a layer.

For the domain with inclusion embedded, ϕ^i is the TDF of the i-th component and Ω^i is the region occupied by the i-th component. For an inclusion embedded composite, the mathematical expression of TDF is:

$$\phi^i(\mathbf{x}) = \sqrt{\left(x - x_o^i\right)^2 + \left(y - y_o^i\right)^2} - r^i\left(\theta^i\right) \tag{10}$$

where θ^i is the angle from the point (x, y) to the center of the i-th component $\left(x_o^i, y_o^i\right)$ in polar coordinate system. $r^i(\theta^i)$ is the distance from the point with the angle of θ^i on $\partial\Omega^i$ to the center $\left(x_o^i, y_o^i\right)$ as shown in Fig. 3.

With the defined TDF, the spatial variation of the shear modulus of a layered composite can be defined based on the Boolean operators [31]:

$$\mu(\mathbf{x}) = \mu^0\big(1 - H(\phi^1(\mathbf{x}))\big) + \sum_{i=1}^{n-1}\big[\mu^i\, H(\phi^i(\mathbf{x}))\big(1 - H(\phi^{i+1}(\mathbf{x}))\big)\big] + \mu^n\, H(\phi^n(\mathbf{x})) \tag{11}$$

where μ^0 is the shear modulus of background. μ^n is the shear modulus of the last layer. $H(z)$ is the Heaviside function. If there is no interface curve C^i, $\mu(\mathbf{x}) = \mu^0$ is a homogeneous domain. However, the TDF definitions of composite with embedded inclusions are different, that is:

$$\boldsymbol{\mu}(\boldsymbol{x}) = \boldsymbol{\mu}^0 \prod_{i=1}^{n} \mathbf{H}\big(\boldsymbol{\phi}^i(\boldsymbol{x})\big) + \sum_{i=1}^{n}\Big(\boldsymbol{\mu}^i\big(1 - \mathbf{H}(\boldsymbol{\phi}^i(\boldsymbol{x}))\big)\Big) \tag{12}$$

As the Heaviside function is a discontinuous function, we will utilize the following function to approximate the Heaviside function so that we can evaluate the spatial gradient of the Heaviside function:

$$H(z) = 0.5 \times \left(1 + \tanh\left(\frac{z}{\varepsilon}\right)\right) \tag{13}$$

where ε is a small constant. With the decrease of ε, the approximation of the Heaviside function is closer to the corresponding analytical expression (see Fig. 4). However, a smaller ε might lead to an extremely large gradient at z approaching to zero. Thus, the small constant ε should be carefully chosen.

Inverse problem

After discussing the TDF, the inverse problem can be stated as: Given n_{meas} surface measured displacement fields, find the optimal geometric vectors $\mathbf{r}^1, \mathbf{r}^2, \ldots, \mathbf{r}^m$ and the shear moduli $\mu^0, \mu^1, \ldots, \mu^m$, such that the following objective function is minimized:

$$\Pi = \frac{1}{2} \sum_{i_{\text{meas}}=1}^{n_{\text{meas}}} \int |\mathbf{u}_i^{\text{com}} - \mathbf{u}_i^{\text{meas}}|^2 \, d\partial\Omega_{i_{\text{meas}}} + \text{Reg}(\mu) \tag{14}$$

[Figure 2 — placeholder]

Bilayer structure illustrating the TDF construction. The bottom layer μ^0 (teal) is bounded below by the reference curve C^0; the top layer μ^1 (pink) is bounded above by the interface curve C^1. Six control points $\mathbf{P}_1^1, \mathbf{P}_2^1, \ldots, \mathbf{P}_6^1$ define the B-spline interface C^1 (orange). Radial distances $r_1^1, r_2^1, \ldots, r_6^1$ connect each control point back to C^0, parameterised along t.

Fig. 2. An example of the nonhomogeneous shear modulus distribution of a bilayer structure.